

Introduction to Cloud Computing and AWS

Prof. Arnab Mandal

Assistant Professor

CSE Department

Content

1. What is Cloud Computing?.....	3
2. Key Benefits of Cloud Computing	4
3. Cloud Service Models	5
4 .Cloud Deployment model.....	10
5. Deployment Model Comparision.....	30
6.Detailed Service Comparison	31
8.The AWS Global Infrastructure.....	33
9. Challenges and Consideration.....	34
12. Key Takeaways.....	35

What is Cloud Computing?

- Cloud computing is the **on-demand delivery** of IT resources over the internet with **pay-as-you-go pricing**. Instead of buying, owning, and maintaining physical data centers and servers, you access technology services, such as computing power, storage, and databases, from a provider like AWS.

Key Benefits of Cloud Computing

- **Agility:** Quickly spin up resources as you need them, reducing the time to deploy applications from weeks to minutes.
- **Elasticity:** Scale resources up or down instantly to meet fluctuating demand.
- **Cost Savings:** Trade capital expense (upfront hardware costs) for variable expense (paying only for what you use).
- **Deploy Globally in Minutes:** Easily deploy your application in multiple regions around the world with just a few clicks.

Cloud Service Models Overview

Three primary service models representing different levels of abstraction and control:

- **IaaS:** Infrastructure as a Service
- **PaaS:** Platform as a Service
- **SaaS:** Software as a Service

Cloud Service Models

Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources such as servers, storage, and networking over the internet on a pay-as-you-go basis. It offers the highest level of control and flexibility, acting as the digital foundation for your IT operations.

How it works: It is like renting an empty plot of land. The provider gives you the space and the utilities (hardware, network, storage), but you are entirely responsible for building the house (installing the OS, middleware, and applications).

Characteristics:

- Eliminates the capital expense of buying physical servers.
- Offers complete control over operating systems, applications, and development frameworks.
- Resources can be scaled up or down instantly based on traffic spikes.

Cloud Service Models

Platform as a Service (PaaS)

PaaS offers a complete, managed cloud environment specifically designed for developing, testing, running, and managing applications. It abstracts the underlying infrastructure so developers can focus purely on writing code.

How it works: It is like renting an unfurnished house. The foundation, plumbing, and electricity are already built and maintained by the landlord; you just need to bring your furniture and decorate (write and deploy your code).

Characteristics:

- Acts as a complete toolkit for developers, providing pre-built tools, libraries, and development environments.
- The cloud provider handles all backend infrastructure, including server provisioning, OS patching, and network routing.
- Automates deployment workflows and allows for easy team collaboration.

Cloud Service Models

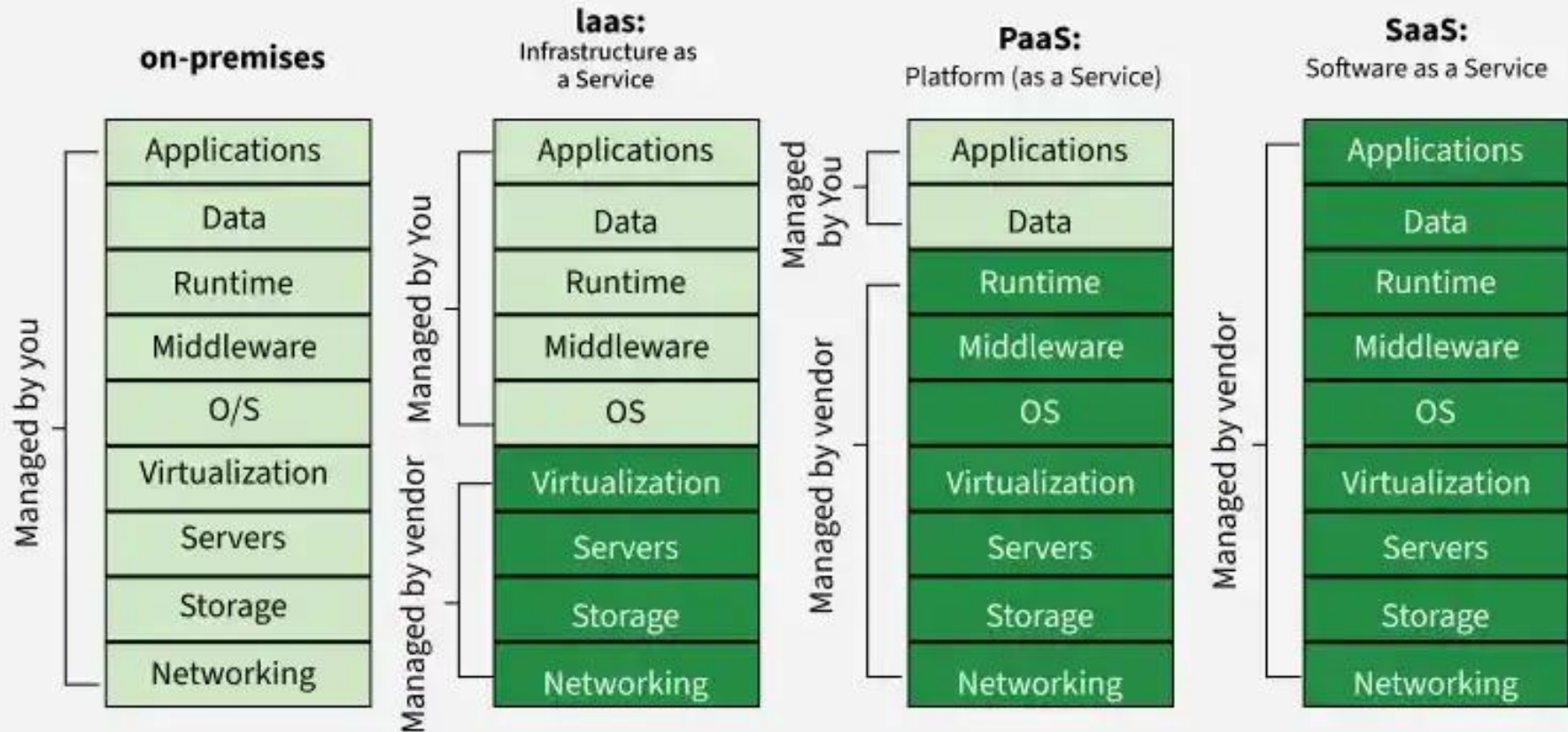
Software as a Service (SaaS)

SaaS is the most user-friendly and widely used cloud model. It delivers fully functional, ready-to-use software applications over the internet, typically on a subscription basis.

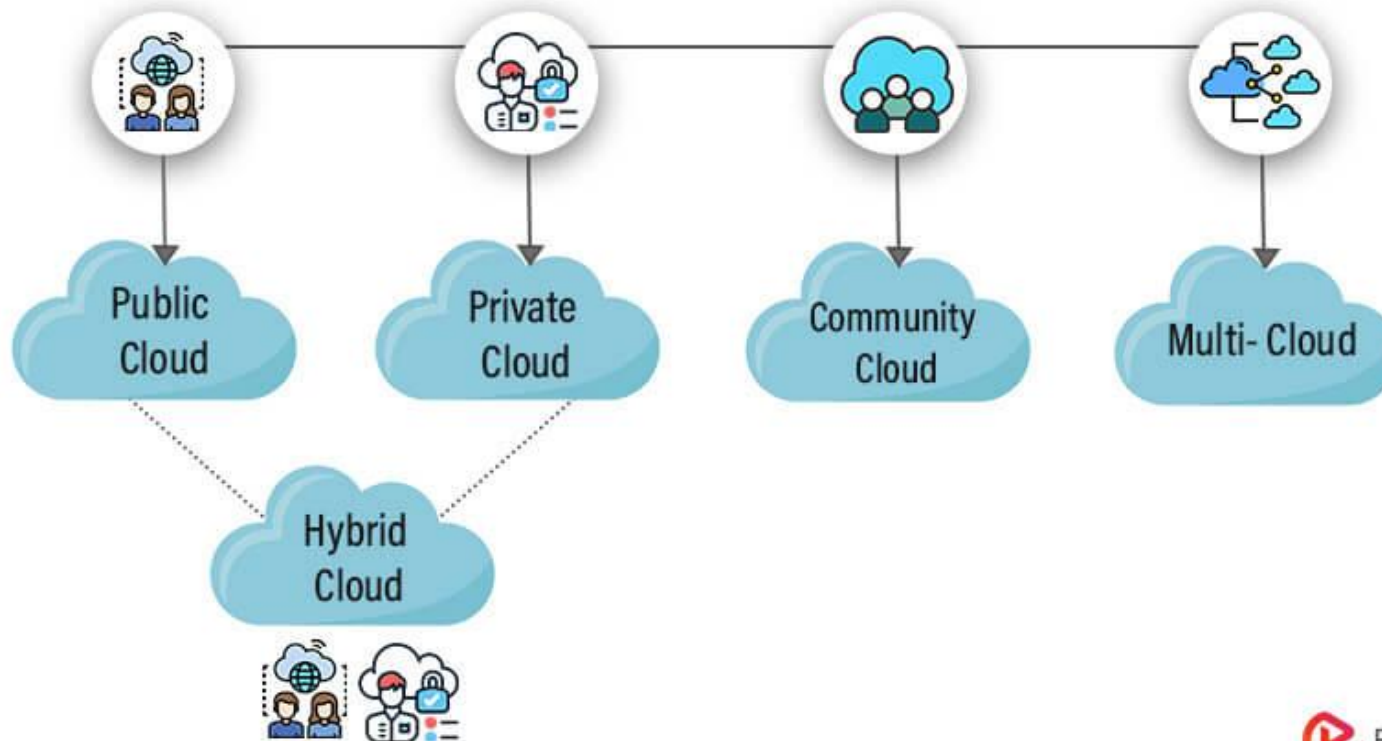
How it works: It is like staying in a fully furnished, serviced hotel room. Everything is provided, maintained, and cleaned for you; you simply walk in and use it.

Characteristics:

- Applications are ready to use immediately directly through a web browser or mobile app.
- The cloud provider handles absolutely everything: infrastructure, software updates, bug fixes, and security maintenance.
- Requires minimal technical expertise to operate.

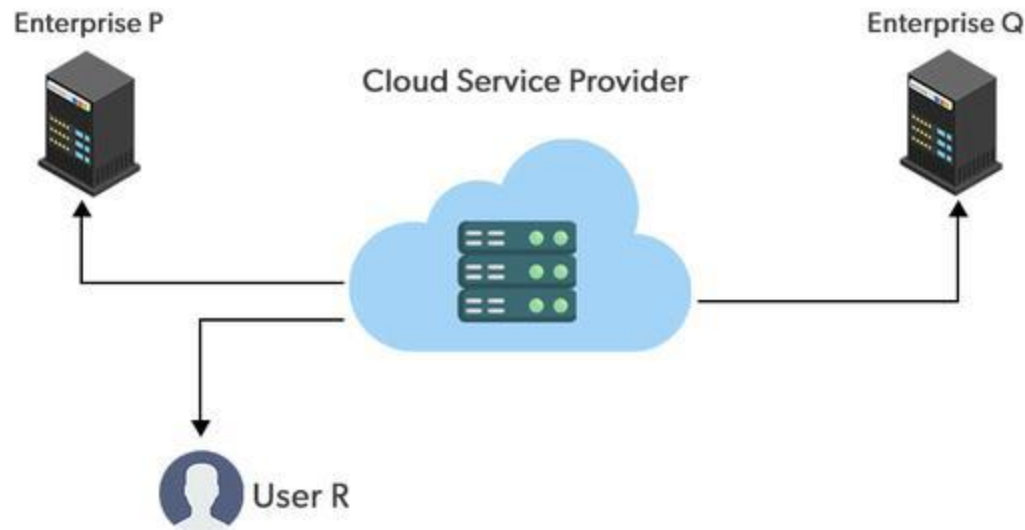


Cloud Deployment Models



Public Cloud

- Public clouds are managed by third parties which provide cloud services over the internet to the public, these services are available as pay-as-you-go billing models.
- They offer solutions for minimizing IT infrastructure costs and become a good option for handling peak loads on the local infrastructure. Public clouds are the go-to option for small enterprises, which can start their businesses without large upfront investments by completely relying on public infrastructure for their IT needs.
- The fundamental characteristics of public clouds are **multitenancy**. A public cloud is meant to serve multiple users, not a single customer. A user requires a virtual computing environment that is separated, and most likely isolated, from other users.



Advantages of Public Cloud

- The following are the advantages of public cloud:
- Public cloud is easily able to scale up and down resources as per the demand of traffic and workload. It facilitates with performance optimization and cost efficiency.
- It works on pay-as-you-go cloud model and helps in resolving the investments needs in hardware and infrastructure reducing overall costs.

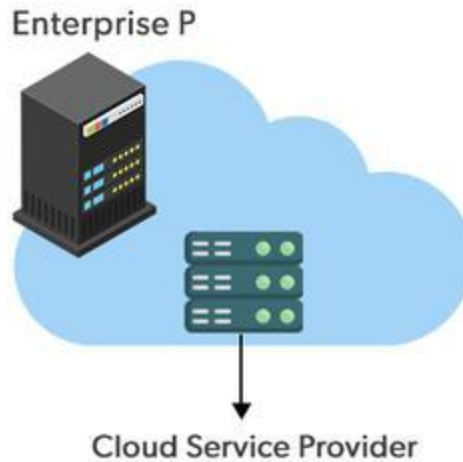
Disadvantages of using Public Cloud

- The following are the disadvantages of Public Cloud:
- It is difficult to trust and maintain data to a third-party provider may raise concerns about control and ownership
- The shared infrastructure of public cloud resources increases the risk of data breaches and unauthorized access. It raises security and privacy concerns.
- Public cloud comes with limited transparency about the underlying infrastructure which may make it challenging to monitor and manage performance effectively.

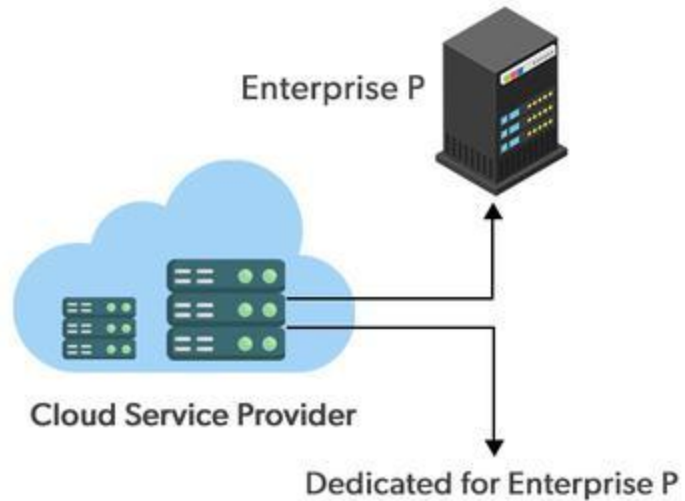
Private cloud

- Private clouds are distributed systems that work on private infrastructure and provide the users with dynamic provisioning of computing resources. Instead of a pay-as-you-go model in private clouds, there could be other schemes that manage the usage of the cloud and proportionally billing of the different departments or sections of an enterprise. Private cloud providers are HP Data Centers, Ubuntu, Elastic-Private cloud, Microsoft, etc.
- **Examples:** VMware vCloud Suite, OpenStack, Cisco Secure Cloud, Dell Cloud Solutions, HP Helion Eucalyptus

On premise Private cloud



Externally hosted Private cloud



Advantages Of Private Cloud

Customer information protection: In the private cloud security concerns are less since customer data and other sensitive information do not flow out of private infrastructure.

Infrastructure ensuring SLAs: Private cloud provides specific operations such as appropriate clustering, data replication, system monitoring, and maintenance, disaster recovery, and other uptime services.

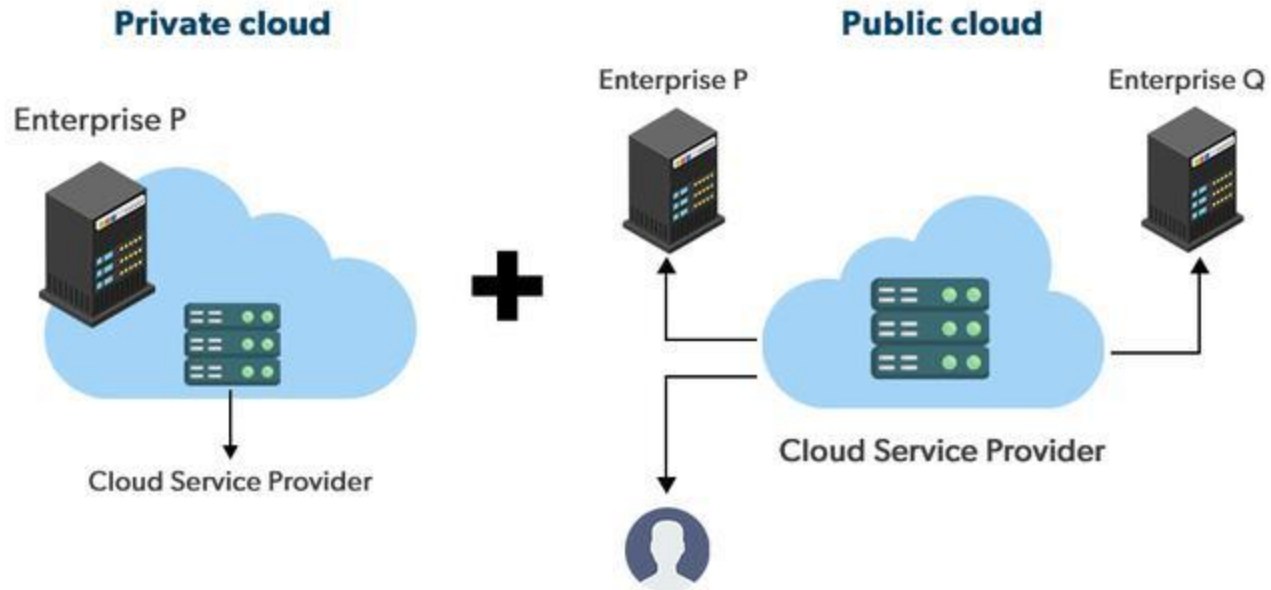
Compliance with standard procedures and operations: Specific procedures have to be put in place when deploying and executing applications according to third-party compliance standards. This is not possible in the case of the public cloud.

Disadvantages Of Private Cloud

- **The restricted area of operations:** Private cloud is accessible within a particular area. So the area of accessibility is restricted.
- **Expertise requires:** In the private cloud security concerns are less since customer data and other sensitive information do not flow out of private infrastructure. Hence skilled people are required to manage & operate cloud services.

Hybrid cloud

- A hybrid cloud is a heterogeneous distributed system formed by combining facilities of the public cloud and private cloud. For this reason, they are also called **heterogeneous clouds**.
- A major drawback of private deployments is the inability to scale on-demand and efficiently address peak loads. Here public clouds are needed. Hence, a hybrid cloud takes advantage of both public and private clouds.
- **Examples:** [AWS Outposts](#), Azure Stack, Google Anthos, IBM Cloud Satellite, Oracle Cloud at Customer



Benefits of Hybrid Cloud Computing

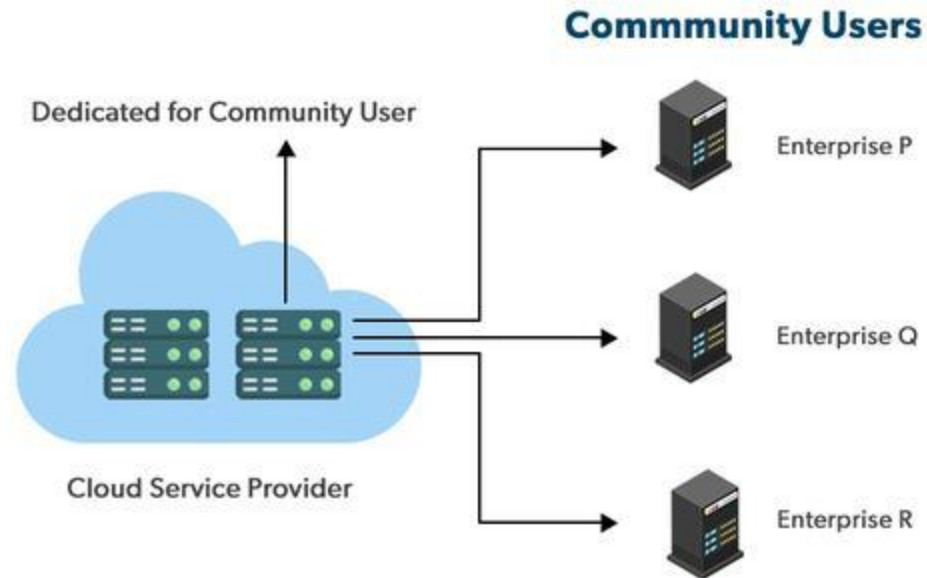
- The following are the advantages of using Hybrid Cloud:
- Hybrid cloud is available at a cheap cost than other clouds because it is formed by a distributed system.
- It works comes up with working fast with lower cost and facilitates in reducing the latency of the data transfer process.
- Most important thing is security. A hybrid cloud is totally safe and secure because it works on the distributed system network.

Disadvantages of Using Hybrid Cloud

- The following are the disadvantages of using Hybrid Cloud:
- It's possible that businesses lack the internal knowledge necessary to create such a hybrid environment. Managing security may also be more challenging. Different access levels and security considerations may apply in each environment.
- Managing a hybrid cloud may be more difficult. With all of the alternatives and choices available today, not to mention the new PaaS components and technologies that will be released every day going forward, public cloud and migration to public cloud are already complicated enough. It could just feel like a step too far to include hybrid.

Community Cloud

- Community clouds are distributed systems created by integrating the services of different clouds to address the specific needs of an industry, a community, or a business sector. But sharing responsibilities among the organizations is difficult.
- In the community cloud, the infrastructure is shared between organizations that have shared concerns or tasks. An organization or a third party may manage the cloud.
- **Examples:** CloudSigma, Nextcloud, Synology C2, OwnCloud, Stratoscale



Advantages of Using Community Cloud

- The following are the advantages of using Community Cloud:
- Because the entire cloud is shared by numerous enterprises or a community, community clouds are cost-effective.
- Because it works with every user, the community cloud is adaptable and scalable. Users can alter the documents according to their needs and requirements.
- Public cloud is less secure than the community cloud, which is more secure than private cloud.
- Thanks to community clouds, we may share cloud resources, infrastructure, and other capabilities between different enterprises.

Disadvantages of using Community Cloud

- The following are the disadvantages of using Community Cloud:
- Not all businesses should choose community cloud.
- Gradual adoption of data
- It's challenging for corporations to share duties.

Applications Of Community clouds

The following are the applications of community clouds:

- **Media industry:** Media companies are looking for quick, simple, low-cost ways for increasing the efficiency of content generation. Most media productions involve an extended ecosystem of partners. In particular, the creation of digital content is the outcome of a collaborative process that includes the movement of large data, massive compute-intensive rendering tasks, and complex workflow executions.
- **Healthcare industry:** In the healthcare industry community clouds are used to share information and knowledge on the global level with sensitive data in the private infrastructure.
- **Energy and core industry:** In these sectors, the community cloud is used to cluster a set of solution which collectively addresses the management, deployment, and orchestration of services and operations.
- **Scientific research:** In this organization with common interests in science share a large distributed infrastructure for scientific computing.

Multicloud

- Multicloud is the use of multiple cloud computing services from different providers, which allows organizations to use the best-suited services for their specific needs and avoid vendor lock-in.
- This allows organizations to take advantage of the different features and capabilities offered by different cloud providers.
- **Examples:** Cloud Foundry, Kubernetes, Apache Mesos, Red Hat OpenShift, Docker Swarm

Advantages of using Multi-Cloud

The following are the advantages of using multi-cloud:

- **Flexibility:** Using multiple cloud providers allows organizations to choose the best-suited services for their specific needs, and avoid vendor lock-in.
- **Cost-effectiveness:** Organizations can take advantage of the cost savings and pricing benefits offered by different cloud providers for different services.
- **Improved performance:** By distributing workloads across multiple cloud providers, organizations can improve the performance and availability of their applications and services.
- **Increased security:** Organizations can increase the security of their data and applications by spreading them across multiple cloud providers and implementing different security strategies for each.

Disadvantages of using Multi-Cloud

The following are the disadvantages of using Multi-Cloud:

- **Complexity:** Managing multiple cloud providers and services can be complex and require specialized knowledge and expertise.
- **Increased costs:** The cost of managing multiple cloud providers and services can be higher than using a single provider.
- **Compatibility issues:** Different cloud providers may use different technologies and standards, which can cause compatibility issues and require additional resources to resolve.
- **Limited interoperability:** Different cloud providers may not be able to interoperate seamlessly, which can limit the ability to move data and applications between them

Deployment Models Comparison

Feature	Public Cloud	Private Cloud	Hybrid Cloud
Ownership	Third-party	Organization	Mixed
Scalability	Elastic	Limited	High
Cost	OpEx	CapEx + OpEx	Blended
Security	Standard	High	Customized

Detailed Service Model Comparison

Aspect	IaaS	PaaS	SaaS
Infrastructure	Provider	Provider	Provider
Platform	Client	Provider	Provider
Control Level	High	Medium	Low
Maintenance	Client	Shared	Provider

What is AWS?

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform. It launched in 2006 and now offers over 200 fully featured services from data centers globally. AWS provides the "Lego bricks" of infrastructure—from basic compute and storage to advanced artificial intelligence and Internet of Things (IoT) tools.

The AWS Global Infrastructure

AWS serves millions of customers by maintaining a massive, resilient physical footprint. Understanding this hierarchy is key to building reliable applications.

- **Regions:** A Region is a physical location in the world where AWS has multiple Availability Zones. Each Region is geographically isolated to ensure maximum fault tolerance and stability.
- **Availability Zones (AZs):** An AZ consists of one or more discrete data centers with redundant power, networking, and connectivity. They are miles apart within a Region to protect against local disasters (like fires or floods) but are connected by ultra-fast fiber.
- **Edge Locations:** These are smaller sites located in major cities globally. They are used by services like **Amazon CloudFront** to cache content closer to end-users, reducing latency.

Challenges and Considerations

- **Security:** Shared responsibility, data protection
- **Vendor Lock-in:** Difficulty switching providers
- **Cost Management:** Runaway spending on public cloud
- **Compliance:** Meeting regulatory requirements
- **Performance:** Network latency and availability

Key Takeaways

- Cloud computing operates on deployment and service models
- Choose deployment model based on security, control, and cost needs
- Service models offer different levels of abstraction and control
- AWS is one of the leading providers with distinct strengths
- Hybrid cloud is becoming the enterprise standard approach

Parul[®]
University

NAAC
GRADE **A++**



<https://paruluniversity.ac.in/>

